



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2020 HSC ASSESSMENT TASK 4

Tuesday, 18 August 2020

General instructions

- Working time – 70 min.
(plus 5 minutes reading time)
- **Commence each new question on a new booklet.**
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.

Class (please ✓)

- ☐ 12MM3A2 – Ms Lee
- ☐ 12MM3B2 – Mr Hwang
- ☐ 12MM3C2 – Ms Sarofim
- ☐ 12MX2A6 – Mr Lin
- ☐ 12MX2B5 – Dr Jomaa
- ☐ 12MX2C4 – Mr Umakanthan

STUDENT NUMBER

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{10}$	$\overline{16}$	$\overline{12}$	$\overline{38}$	

Question 1	(10 Marks)	Commence a NEW booklet.	Marks
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- (a) A projectile is launched from the top of an 50 metre building that stands on level ground. The particle's initial velocity vector is $30\vec{i} + 15\vec{j}$. (Letting the ground level of the building be the origin)

Assuming acceleration due to gravity is $10ms^{-2}$

- | | |
|---|----------|
| i. Express its velocity at time t in component vector form ($\vec{v} = x\vec{i} + y\vec{j}$). | 2 |
| ii. Express its displacement at time t in component vector form. ($\vec{r} = x\vec{i} + y\vec{j}$). | 2 |
| iii. Find: | |
| A. the initial speed of the projectile, | 2 |
| B. the position vector of the projectile after 2 seconds, | 2 |
| C. the position vector of the projectile when it reaches the ground. | 2 |

Question 2 (16 Marks)

Commence a NEW booklet.

Marks

(a) Use the reverse chain rule to find:

i. $\int \sin x \cos^5 x \, dx$ **2**

ii. $\int \cot 7x \, dx$ **2**

(b) Find the exact value of each definite integral.

i. $\int_2^3 \frac{3x}{\sqrt{x^2 - 2}} \, dx$ **2**

ii. $\int_0^{\frac{\pi}{2}} 3 \sin x \, e^{\cos x} \, dx$ **3**

(c) i. Sketch the curves $y^3 = x$ and $32y = x^2$ **2**

ii. Show that the points of intersection of the curves are at $x = 0$ and $x = 8$ algebraically. **1**

iii. Show that the area formed between these curves is given by **1**

$$\int_0^8 \left(x^{\frac{1}{3}} - \frac{x^2}{32} \right) dx$$

iv. Find the volume of the solid when this region is rotated about the x -axis. **3**

Question 3 (12 Marks)

Commence a NEW booklet.

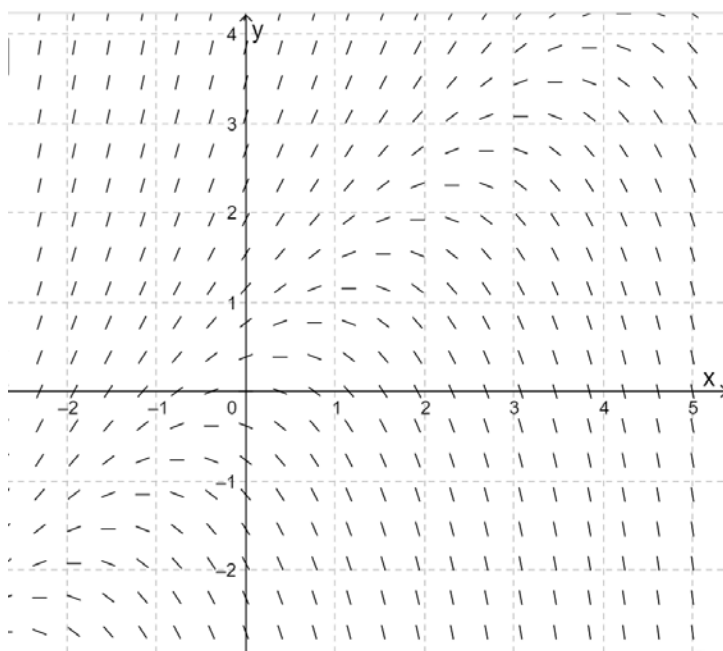
Marks

(a) For differential equation $y' = \frac{5}{6x^2 + x - 1}$:

i. Show that $y' = \frac{3}{3x-1} - \frac{2}{2x+1}$. 1

ii. Find the solution to the differential equation; given that when $x = -\frac{1}{4}$, $y = \frac{1}{4}$. 2

(b) The direction field for the differential equation $\frac{dy}{dx} + x - y = 0$ is shown below. A solution to this differential equation that includes (2,-1) could also include 1



- A. (3,-0.2)
- B. (3,-2)
- C. (-1,-0.2)
- D. (-1,-2)

(c) A farmer releases 20 rabbits onto a remote island off the coast of Sydney. The island is believed to support at most 5900 rabbits. The growth rate of rabbit population P is $\frac{dP}{dt} = kP(5900 - P)$, $P(0) = 20$ with $k > 0$ and t measured in years

i. Show that $\frac{1}{P(5900 - P)} = \frac{1}{5900} \left(\frac{1}{P} + \frac{1}{5900 - P} \right)$ 1

ii. Find the equation for the population P at time t years after the release. 3

iii. Find the value of k correct to four significant figures, if the population of the rabbits reaches 3200 after 4 years. 3

iv. What will the population be after 9 years, correct to the nearest whole number? 1

End of Examination

Suggested Solutions

Question 1

(a) i. (2 marks)

Initially $x = 0, y = 50, \dot{x} = 30, \dot{y} = 15$

$\ddot{x} = 0, \dot{x} = C_1$, when $t = 0$,

$\dot{x} = 30$

$\ddot{y} = -10, \dot{y} = -10t + C_2$, when $t = 0, \dot{y} = 15 = -10(0) + C_2 \quad C_2 = 15$

$\dot{y} = -10t + 15$

$$\vec{v} = 30\vec{i} + (15 - 10t)\vec{j}$$

ii. (2 marks) $\dot{x} = 30, x = 30t + C_3$

using initial condition, $t = 0, x = 0 \therefore C_3 = 0$

$x = 30t$

$\dot{y} = -10t + 15, y = -5t^2 + 15t + C_4$

using initial condition, $t = 0, y = 50 \therefore C_4 = 50$

$y = -5t^2 + 15t + 50$

$$\vec{r} = 30t\vec{i} + (50 + 15t - 5t^2)\vec{j}$$

iii. A. (2 marks)

$$\vec{v} = 30\vec{i} + 15\vec{j}.$$

$$v^2 = 30^2 + 15^2$$

$$v = \sqrt{30^2 + 15^2} = 15\sqrt{5}ms^{-1}$$

B. (2 marks)

$$\vec{r} = 30t\vec{i} + (50 + 15t - 5t^2)\vec{j}$$

when $t = 2$,

$$\vec{r} = 60\vec{i} + 60\vec{j}$$

C. (2 marks)

when $y = 0$

$$0 = -5t^2 + 15t + 50$$

$$t^2 - 3t - 10 = 0$$

$$t = 5 \text{ as } t > 0$$

$$\vec{r} = 150\vec{i}$$

Question 2

(a) i. (2 marks)

$$\int \sin x \cos^5 x \, dx$$

$$\text{Let } u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$dx = -\frac{du}{\sin x}$$

$$= -\int u^5 \, du$$

$$= -\frac{u^6}{6} + C$$

$$= -\frac{1}{6} \cos^6 x + C$$

ii. (2 marks)

$$\int \cot 7x \, dx$$

$$\text{Let } u = 7x$$

$$\frac{du}{dx} = 7$$

$$dx = \frac{du}{7}$$

$$\frac{1}{7} \int \cot u \, du$$

$$= \frac{1}{7} \int \frac{\cos u}{\sin u} \, du$$

$$\text{Let } v = \sin u$$

$$\frac{dv}{du} = \cos u$$

$$du = \frac{dv}{\cos u}$$

$$\frac{1}{7} \int \frac{1}{v} \, dv$$

$$= \frac{1}{7} \ln(\sin u) + C$$

$$= \frac{1}{7} \ln(\sin 7x) + C$$

(b) i. (2 marks)

$$\int_2^3 \frac{3x}{\sqrt{x^2 - 2}} dx$$

$$\text{Let } u = x^2 - 2$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$\text{upper bound } u = 2^2 - 2 = 2$$

$$\text{lower bound } u = 3^2 - 2 = 7$$

$$\frac{3}{2} \int_2^7 u^{-\frac{1}{2}} dx = \frac{3}{2} \times 2\sqrt{u} \Big|_2^7$$

$$= 3(\sqrt{7} - \sqrt{2})$$

ii. (3 marks)

$$\int_0^{\frac{\pi}{2}} 3 \sin x e^{\cos x} dx$$

$$\text{Let } u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$dx = -\frac{du}{\sin x}$$

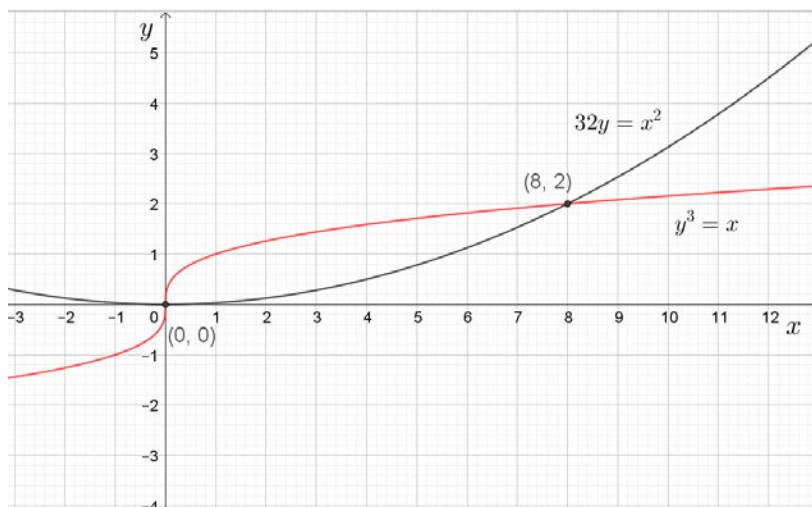
$$\text{upper bound } u = 1$$

$$\text{lower bound } u = 0$$

$$-3 \int_1^0 e^u dx = 3 \int_0^1 e^u dx = 3 \times e^u \Big|_0^1$$

$$= 3(e - 1)$$

(c) i. (2 marks)



ii. (1 mark)

$$y = \frac{x^2}{32}$$

$$\left(\frac{x^2}{32}\right)^3 = x$$

$$\frac{x^6}{2^{15}} = x$$

$$x^6 = 2^{15}x$$

$$x(x^5 - 2^{15}) = 0$$

$$(0, 0), (2^3 = 8, 2)$$

iii. (1 mark)

$$y = \frac{x^2}{32}, y = \sqrt[3]{x}$$

$$y = \sqrt[3]{x} \text{ is above } y = \frac{x^2}{32} \text{ as shown in (i)}$$

$$\text{Area} = \int_0^8 \left(x^{\frac{1}{3}} - \frac{x^2}{32} \right) dx$$

iv. (3 marks)

$$V_1 = \pi \int_0^8 x^{\frac{2}{3}} dx = \pi \times \frac{3x^{\frac{5}{3}}}{5} \Big|_0^8 = \frac{96\pi}{5}$$

$$V_2 = \pi \int_0^8 \frac{x^4}{2^{10}} dx = \pi \times \frac{x^5}{5 \times 2^{10}} \Big|_0^8 = \frac{32\pi}{5}$$

$$V = \frac{96\pi}{5} - \frac{32\pi}{5} = \frac{64\pi}{5} u^3 \approx 40.21u^3 (2d.p)$$

Question 3

(a) i. (1 mark)

$$y' = \frac{3}{3x-1} - \frac{2}{2x+1} = \frac{3(2x+1) - 2(3x-1)}{(3x-1)(2x+1)}$$

$$= \frac{5}{(3x-1)(2x+1)} \text{ as required}$$

ii. (2 marks)

$$y = \int \frac{3}{3x-1} - \frac{2}{2x+1} dx$$

$$y = \ln|3x-1| - \ln|2x+1| + C$$

$$y = \ln \left| \frac{3x-1}{2x+1} \right| + C$$

$$\text{sub } x = -\frac{1}{4}, y = \frac{1}{4}$$

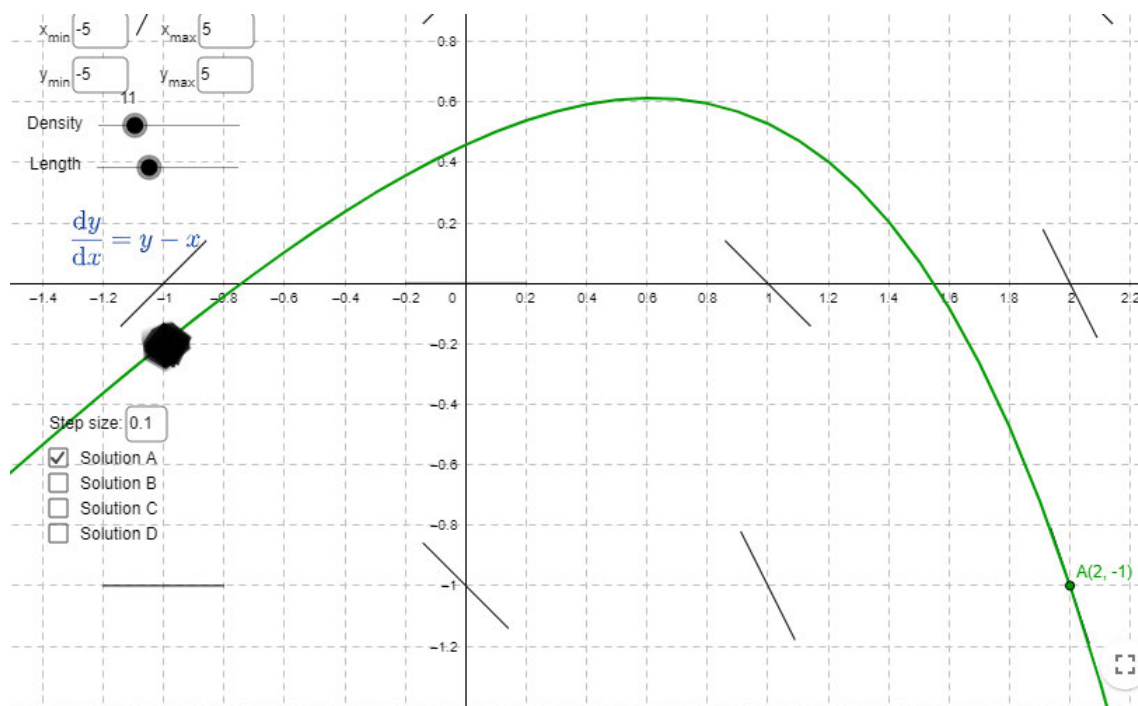
$$C = \frac{1}{4} - \ln \left| -\frac{7}{2} \right|$$

$$\therefore y = \ln \left| \frac{3x-1}{2x+1} \right| + \frac{1}{4} - \ln \frac{7}{2}$$

(b) (1 mark)

Draw the solution, $(-1, -0.2)$ is on the line.

Answer is C



(c) i. (1 mark)

$$\begin{aligned} RHS &= \frac{1}{5900} \left(\frac{1}{P} + \frac{1}{5900 - P} \right) \\ &= \frac{1}{5900} \left(\frac{5900 - P + P}{P(5900 - P)} \right) \\ &= \frac{1}{P(5900 - P)} \text{ as required} \end{aligned}$$

ii. (3 marks)

$$\begin{aligned} \frac{dP}{dt} &= kP(5900 - P) \\ \int \frac{1}{P(5900 - P)} dP &= \int k dt \\ \int \frac{1}{P} + \frac{1}{5900 - P} dP &= 5900 \int k dt \\ \ln |P| - \ln |5900 - P| &= 5900kt + C \\ \ln \left| \frac{P}{5900 - P} \right| &= 5900kt + C \\ \left| \frac{P}{5900 - P} \right| &= Ae^{5900kt} \text{ where } A = e^C > 0 \\ \frac{P}{5900 - P} &= Ae^{5900kt} \text{ where } A = \pm e^C \\ P &= 5900Ae^{5900kt} - PAe^{5900kt} \\ P(1 + Ae^{5900kt}) &= 5900Ae^{5900kt} \\ P &= \frac{5900Ae^{5900kt}}{1 + Ae^{5900kt}} = \frac{5900}{1 + Be^{-5900kt}} \text{ where } B = A^{-1} \\ t = 0, P &= 20 \\ 20 &= \frac{5900}{1 + B} \\ B &= 294 \\ \therefore P &= \frac{5900}{1 + 294e^{-5900kt}} \end{aligned}$$

iii. (3 marks)

$$\begin{aligned} t = 4, P &= 3200 \\ 3200 &= \frac{5900}{1 + 294e^{(-4 \times 5900)k}} \\ k &= \frac{\ln \frac{5900 - 3200}{294 \times 3200}}{-4 \times 5900} \\ k &\approx 2.480 \times 10^{-4} (4s.f) \end{aligned}$$

iv. (1 mark)

$$t = 9$$

$$P = \frac{5900}{1 + 294e^{(-5900 \times 9)k}} = 5896.695 \dots$$

5897 rabbits